

Spatial Population Dynamics: Fertility, Migration, and Policies

Zi Wang¹ Xican Xi² Anson Zhou³

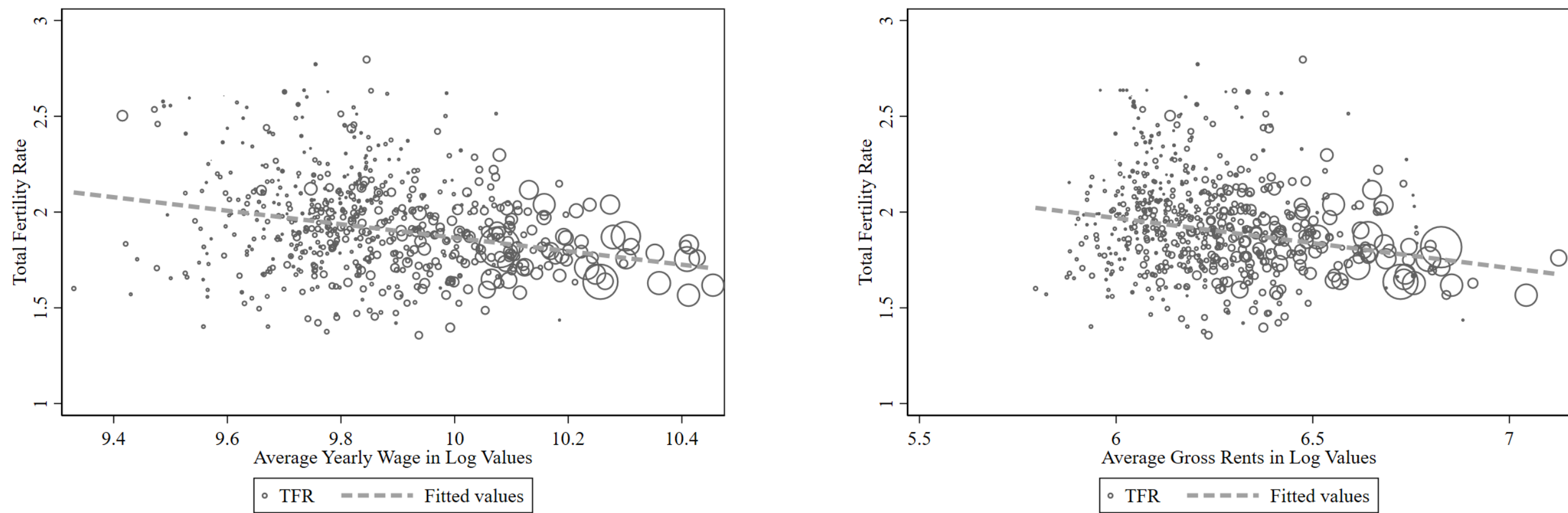
¹Hong Kong Baptist University

²Fudan University

³The University of Hong Kong

Motivation

- U.S. TFR fell from 2.1 (1990) to 1.6 (2023).
- Across 741 U.S. commuting zones (CZs), fertility is lower where wages and rents are high.
- We model fertility as a spatial equilibrium outcome, jointly with location, housing, and children.



Circle size = population. Data: 2000 U.S. Census (IPUMS).

Research Questions

- How do regional policies affect welfare and population growth?
- How do housing supply constraints affect aggregate fertility?
- Can transfers improve the welfare–fertility trade-off?

Model Framework

Spatial OLG model with N regions, trade, migration, and endogenous fertility.

- Households** choose location (Fréchet η), consumption C , housing h , and children n .

$$\text{Utility } V_i = [\alpha^{1/\gamma} C_i^{(\gamma-1)/\gamma} + (1-\alpha)^{1/\gamma} \underbrace{[\beta^{1/\rho} h_i^{(\rho-1)/\rho} + (1-\beta)^{1/\rho} n_i^{(\rho-1)/\rho}]^{\frac{\rho}{\rho-1}}}_{\text{housing-fertility composite}}^{\frac{\rho-1}{\rho}}]^{\frac{\gamma}{\gamma-1}}$$
$$\text{s.t. } P_{i,t} C_{i,t} + r_{i,t} h_{i,t} = w_{i,t} (1 - \chi_{i,t} n_{i,t}) + T_{i,t}$$

- Production:** CES aggregation (σ), agglomeration ψ , iceberg trade costs τ_{in} .
- Housing supply:** region-specific inverse elasticity ξ_i (Saiz, 2010).
- Equilibrium:** labour and housing markets clear; fertility and migration determine population dynamics.

Key elasticities: γ (consumption vs. family composite), ρ (housing vs. children).

Theoretical Results

1. Local price elasticities. Define $\varepsilon_{n,w} \equiv \partial \log n_i / \partial \log w_i|_{T_i}$ and $\varepsilon_{n,r} \equiv \partial \log n_i / \partial \log r_i|_{T_i}$. With $\rho < \gamma$, housing and children are strong complements:

$$\varepsilon_{n,w} = \underbrace{-\rho - (\gamma - \rho)\tilde{\omega}_i^n - (1 - \gamma)(1 - \omega_i^C)\tilde{\omega}_i^n}_{\text{substitution effect}} + \underbrace{\frac{w_i}{w_i + T_i}}_{\text{income effect}}$$
$$\varepsilon_{n,r} = \underbrace{\tilde{\omega}_i^h [-(\gamma - \rho) - (1 - \gamma)(1 - \omega_i^C)]}_{\text{price effect}} \leq 0 \quad (\rho \leq \gamma < 1)$$

where ω_i^C is the consumption share and $\tilde{\omega}_i^h, \tilde{\omega}_i^n$ are within-family shares. Higher wages or rents reduce fertility when the substitution effect dominates; the rent effect is unambiguously negative.

2. General equilibrium propagation. Shocks propagate through goods trade, migration, and housing markets, generating dispersed fertility responses.

3. Endogenous fertility stabilizes dynamics. Positive local shocks raise wages/rents and lower fertility, dampening long-run population responses relative to exogenous fertility:

$$\hat{L}_j^{SS}|_{\text{endog } n} < \hat{L}_j^{SS}|_{\text{exog } n}.$$

4. Aggregate fertility decomposition.

$$d\bar{n} = \underbrace{\sum_i s_i dn_i}_{\text{intensive margin}} + \underbrace{\sum_i (n_i - \bar{n}) ds_i}_{\text{migration margin}}.$$

Reallocation toward low-fertility cities lowers aggregate fertility; reducing costs in gaining cities aligns margins.

5. Welfare vs. growth transfers. Welfare planner concentrates population in large, productive, low-fertility cities; growth planner spreads population toward smaller, higher-fertility areas. Housing reform narrows this trade-off.

Calibration & Key Parameters

- 741 U.S. CZs**, 2000 Census; pre-determined $\eta, \sigma, \psi, \xi_i$.
- SMM** disciplines $\gamma, \rho, \alpha, \beta$ using four moments, incl. the IV housing–fertility moment below.

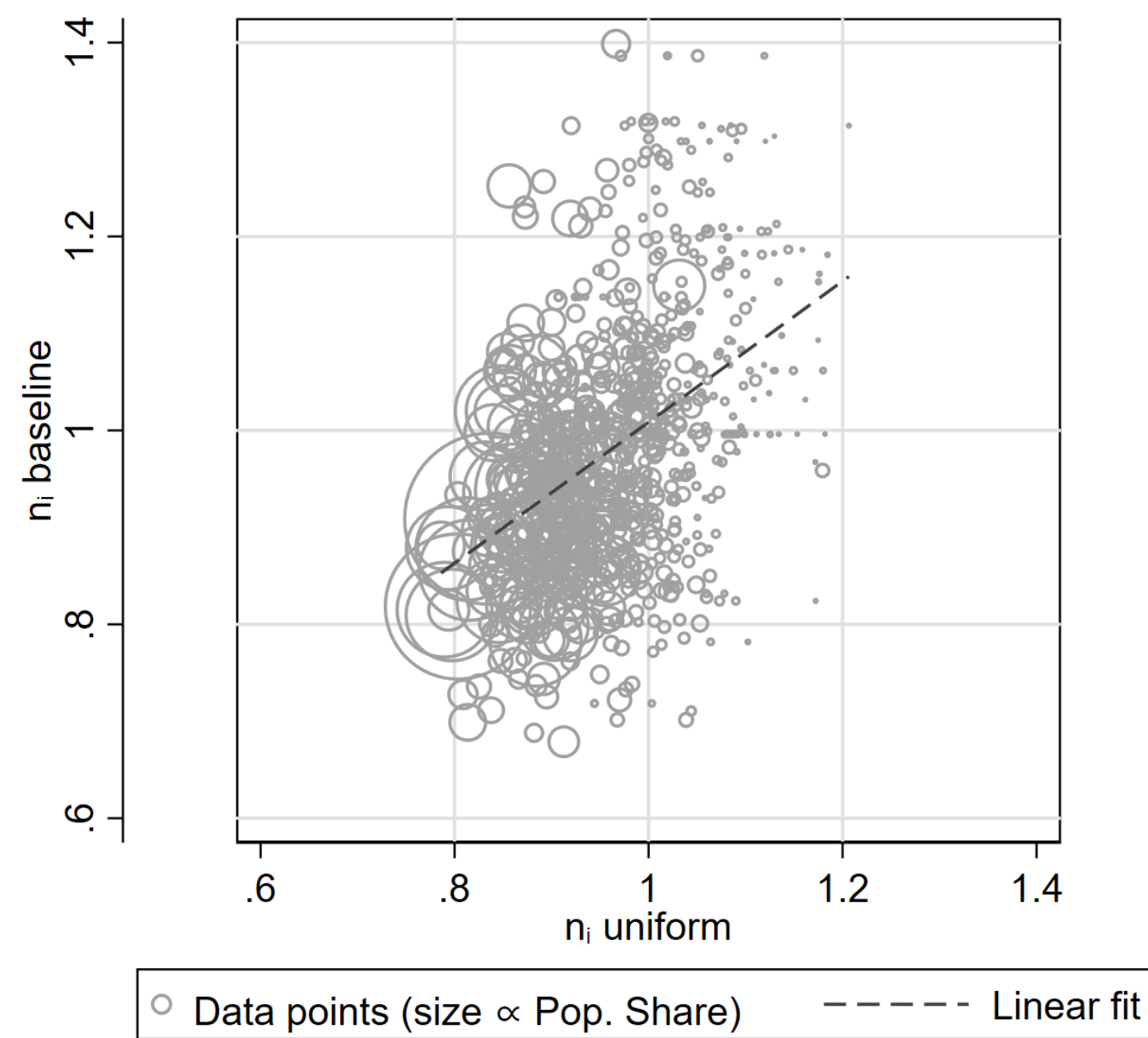
IV housing–fertility moment: $\ln \text{TFR}_{jt} = \beta_1 \text{NB}_{jt} + \beta_2 \text{NB}_{jt} \times \text{Own}_j + \text{controls}$; instrument $\text{Elasticity}_j \times \text{hp}_t$. At mean homeownership (0.66), marginal effect = 0.51 targets ρ .

	OLS	2SLS
NB	−1.398** (0.573)	−1.220*** (0.357)
NB×Own	2.161** (0.918)	2.619*** (0.679)
Target at Own=0.66		0.51
Cragg–Donald F		63.30

Parameter	Value	Interpretation
γ	0.76	Consumption & family composite are complements
ρ	0.38	Housing & children are strong complements
α	0.67	Consumption share
β	0.58	Housing share within family composite

Model Fit

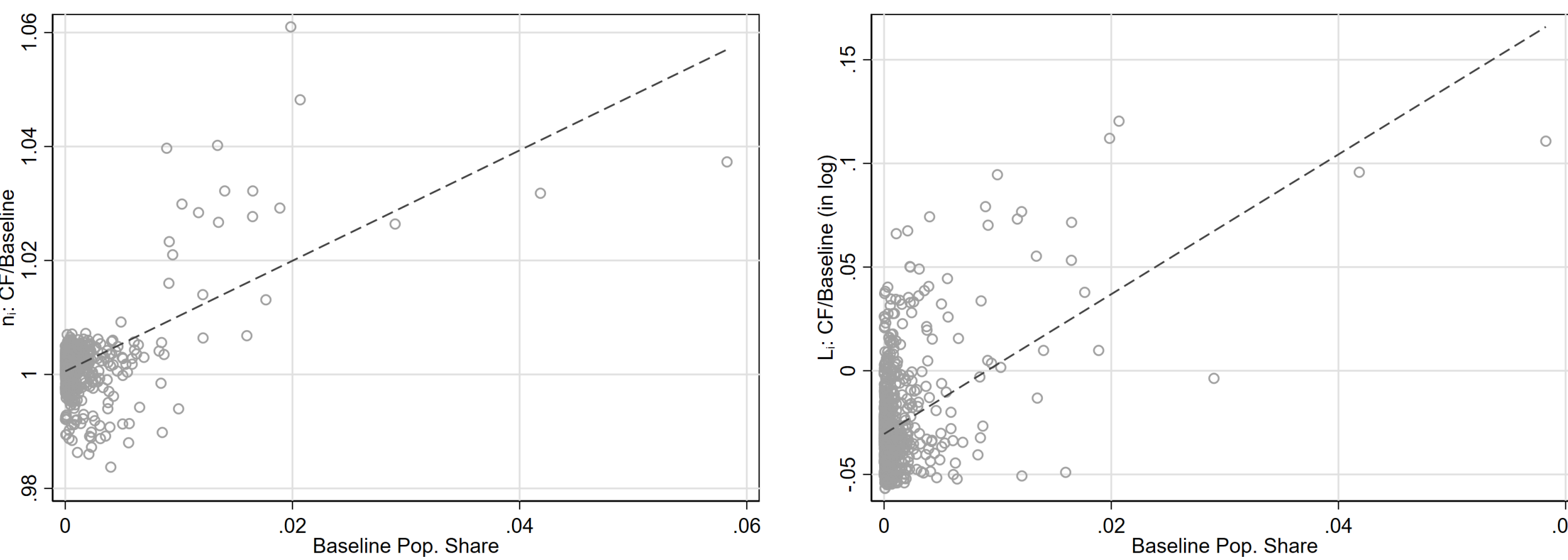
Under uniform parenting cost, the model explains 16% of cross-sectional fertility variation and reproduces negative correlations with wages, population, and rents.



Fertility under uniform $\bar{\chi}$: slope 0.73 (s.e. 0.06); $R^2 = 0.165$. Housing-rent fit: $R^2 = 0.63$.

Counterfactual 1: Relaxing Housing Supply Regulations

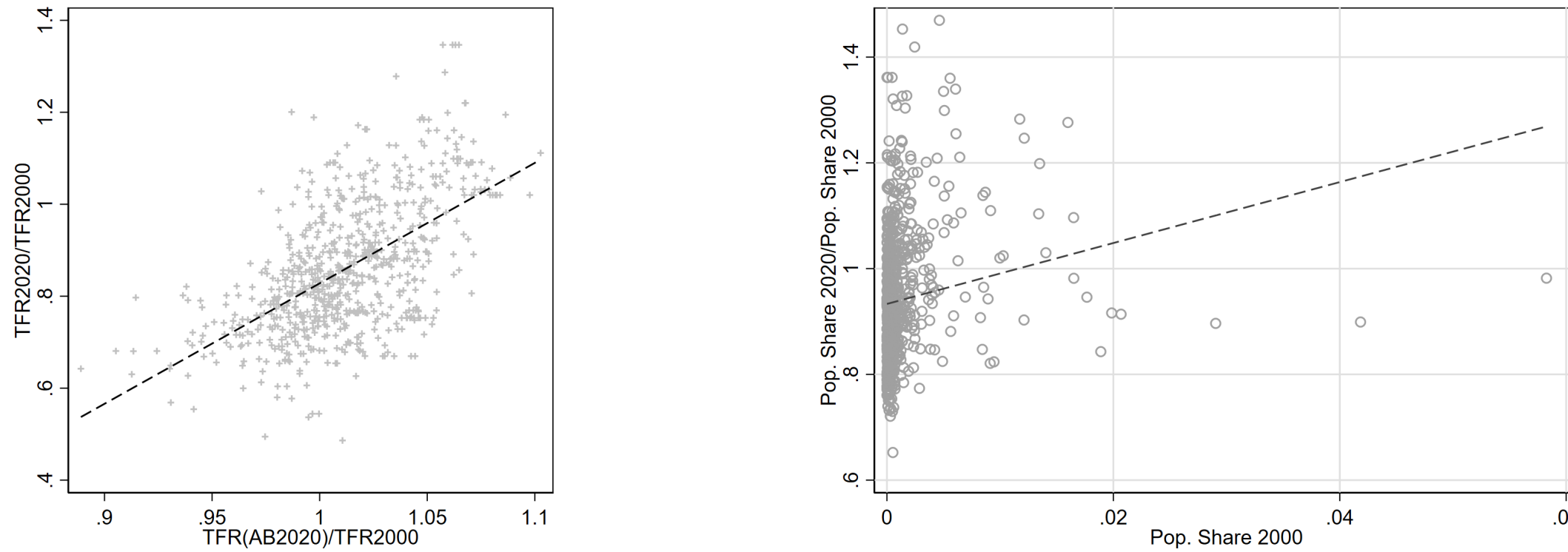
- Experiment:** Set housing regulation in the 22 largest CZs to the median.
- Aggregate:** steady-state population growth g rises from 0.9188 to 0.9281; per-capita welfare +1.83%.



- Mechanism:** lower rents in productive cities raise fertility where inflows occur; margins reinforce.
- Welfare decomposition: housing 80.6%, consumption 11.2%, fertility 8.2%.

Counterfactual 2: Fertility Accounting (2000–2020)

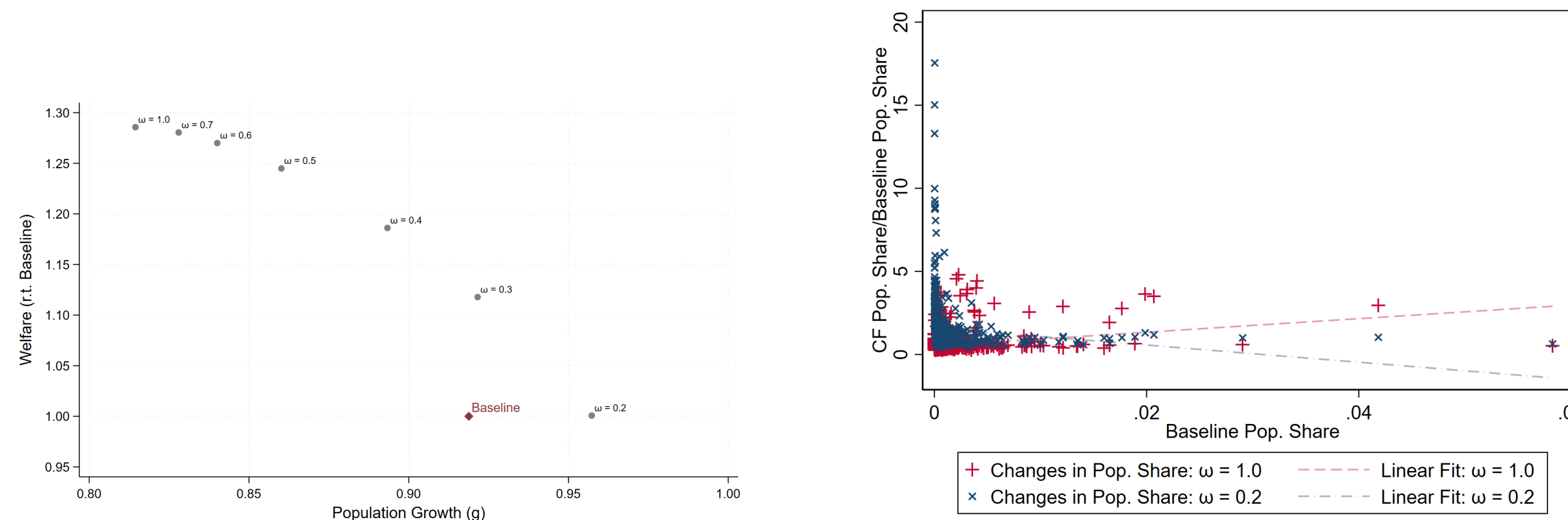
- Update (A_i, B_i) or $\bar{\chi}_i$ from 2000 to 2020 values.
- Productivity & amenities explain 36.6% of cross-CZ fertility changes.
- Population shifted toward large, low-fertility CZs, amplifying the spatial fertility gradient.



Counterfactual 3: Optimal Interregional Transfers

Social planner chooses transfers t_n to maximise $\omega \log W + (1 - \omega) \log g$, with $T_n = \frac{t_n}{L_n} \sum r_k H_k$.

- $\omega = 1$ (welfare): concentrates population in few large CZs, welfare +30%, g falls to 0.81.
- $\omega = 0.2$ (growth): spreads population to smaller, low-cost CZs, g rises to 0.96, welfare near baseline.



Left: Welfare–population frontier. Right: Population shares under $\omega = 1$ (black) vs $\omega = 0.2$ (red).

Conclusions & Policy Implications

- Housing and children are strong complements $\Rightarrow$ housing costs powerfully affect fertility.
- Housing supply reform** in productive cities can raise aggregate fertility and welfare—the efficiency–demography trade-off is not inevitable.
- Aggregate fertility depends on both local responses and spatial reallocation.
- Optimal transfers weigh welfare vs. growth; place-based policies have demographic consequences.